Sentinel-1-Only Land Use and Land Cover Segmentation: Generalisation Limits, Radar-Separable Classes and a Monthly Change-Monitoring Prototype

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***Abstract*—Optical satellites cannot observe the ground through clouds, which limits monthly land use and land cover (LULC) mapping in many regions. This work studies how far synthetic aperture radar alone can replace them. A global dataset of Sentinel-1 monthly composites with Dynamic World labels was built for 40 study areas on six continents, split by region and by time (training 2023–24, testing on nine unseen regions in 2025–26). Ten deep-learning models were trained with an identical protocol: a baseline CNN, a U-Net, DeepLabV3+ with ResNet50, six DeepLabV3+ variants with deeper or more efficient encoders and an atrous-encoder design, and a multi-temporal DeepLabV3+. Spatial context and ImageNet pretraining improve the test mIoU from 0.20 to 0.30, but larger encoders and temporal input do not improve it further, although the same models reach 0.56–0.58 mIoU on the training regions in the test period. The analysis shows that most errors fall into two class pairs that C-band backscatter cannot separate physically, bare ground versus snow/ice and cropland versus rangeland. Merging them into a radar-separable six-class scheme raises the overall accuracy of the same predictions from 0.51 to 0.72. A prototype derives twelve monthly maps, class shares and changes for any location; a case study of Munich agrees with the independent Esri land-cover map on 87.9 % of the pixels and illustrates how seasonal effects and systematic classification errors affect the calculated change.**

***Index Terms*—Sentinel-1, synthetic aperture radar, land use and land cover, semantic segmentation, DeepLabV3+, domain generalisation, change detection, Google Earth Engine**

I. Introduction

Land use and land cover maps are a basic input for monitoring urban growth, deforestation, agriculture and flooding. Recent global products such as Dynamic World [1], the Esri 10 m Annual LULC map [2] and ESA WorldCover [3] are derived from Sentinel-2 optical imagery with deep learning. Optical sensors, however, see nothing through clouds or at night, so in the tropics, in monsoon regions and in northern winters a monthly map often cannot be produced at all. Synthetic aperture radar (SAR) such as Sentinel-1 [4] transmits its own microwave pulse and observes the surface independently of weather and illumination. The price is a signal that is harder to interpret: backscatter depends on surface roughness, geometry and moisture rather than on colour [5], and it is corrupted by speckle.

This work investigates LULC segmentation that uses Sentinel-1 as its *only* input, and the use of such maps to quantify monthly change. The study is organised around three questions: (i) which network design is suitable, (ii) how well a model trained on some regions transfers to unseen regions and to a later period, and (iii) what limits the accuracy and how reliable the derived change statistics are. The contributions are:

- a global Sentinel-1/Dynamic World dataset of 40 study areas on six continents with a split by region and by time, built entirely in Google Earth Engine [6] and published for

reuse;

- a controlled comparison of ten segmentation models, from a small CNN to DeepLabV3+ variants with up to 75 M parameters and a multi-temporal input, all trained with the same protocol and diagnosed for under- and overfitting;
- evidence that the limiting factors are geographic diversity of the training data and the optical origin of the class scheme, not network capacity, and a radar-separable six-class evaluation that raises the overall accuracy from 0.51 to 0.72 without retraining;
- a prototype that maps any location over twelve months, reports class shares and changes in percentage points, flags unreliable months and compares the result with an independent map, demonstrated in a case study of Munich.

II. Related Work

SAR land-cover mapping. Curated benchmarks such as SEN12MS [7] pair Sentinel-1 and Sentinel-2 patches with land-cover labels for deep learning and data fusion. Multi-temporal Sentinel-1 series are effective for crop-type mapping because the phenological cycle changes backscatter over the season [8]. The physics of the main confusions are well known: wet snow strongly absorbs C-band radiation and appears dark, whereas dry snow is largely transparent [9].

Segmentation networks. U-Net [10] combines an encoder-decoder with skip connections. DeepLab [11] introduced atrous (dilated) convolution and atrous spatial pyramid pooling (ASPP) to enlarge the receptive field without losing resolution; DeepLabV3 [12] refined ASPP with image-level features, and DeepLabV3+ [13] adds a decoder for sharper boundaries. Common encoders are ResNet [14] and Xception [15], typically pretrained on ImageNet [16].

Generalisation and change. Models trained in some regions degrade in others because of domain shift [17]. For change estimation, classification errors in two maps compound, and good practice requires accuracy assessment of the change itself [18].

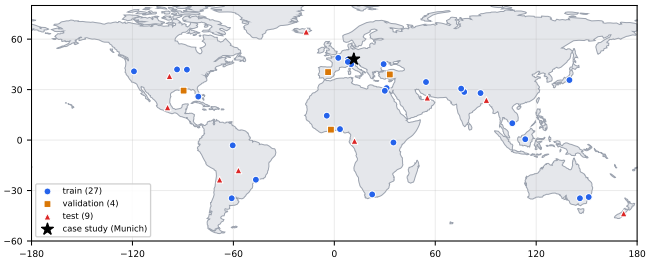


Fig. 1. The 40 study areas. Training, validation and test areas are disjoint; the Munich case study lies outside all of them (nearest study area 260 km away).

TABLE I
Data splits (every second month, offset per year so that all calendar months occur)

Split	Regions	Areas	Period (months)	Patches
Training	train	26*	2023–24 (12)	14 952
Validation	val	4	2023–24 (12)	1 808
Test	unseen	9	2025–26 (11)	4 536
Test temporal	train	26	2025–26 (6)	7 041

*27 planned; Everest lost to the Earth Engine quota (Section III-D).

III. Study Area and Data

A. Study areas and splits

Forty square study areas of 20.48×20.48 km (2048 pixels at 10 m, local UTM grid) were placed on six continents to cover urban, cropland, forest, wetland, rangeland, bare and snow/ice landscapes (Fig. 1). The split is by region *and* by time (Table I): no test region contributes to training or model selection, and the test period follows the training period. A fourth set, `test_temporal`, applies the models to the training regions in the test period and separates the effect of new places from that of new dates. Four initially planned areas (Sahara, Okavango, Gobi, Kazakh steppe) had no Sentinel-1 interferometric wide-swath (IW) acquisition in 2023–24, a consequence of the reduced acquisition plan after the failure of Sentinel-1B in December 2021, and were replaced by areas of the same landscape type.

B. Sentinel-1 input and preprocessing

The input is the Sentinel-1 ground-range-detected collection of Earth Engine

(COPERNICUS/S1_GRD; IW mode, VV and VH, 10 m). In this collection every scene is already processed with the ESA Sentinel-1 toolbox: orbit-file correction, GRD border-noise removal, thermal-noise removal, radiometric calibration to σ^0, terrain correction with the SRTM 30 m elevation model (ASTER above 60° latitude) and conversion to decibels. On top of that, the following steps were applied (Table II):

Orbit pass. Backscatter depends on look direction and incidence angle, so ascending and descending scenes of the same place differ. Each study area uses only the pass with more scenes in its period; mixing passes would create apparent month-to-month change.

Residual border noise and speckle. Pixels with VV below -30 dB are masked as residual border noise or no-return areas. Each scene is filtered in linear power with the Lee filter [19],

$$\hat{x} = \bar{x} + W\,(x - \bar{x}), \quad W = \mathrm{clip}_{[0,1]}\left(\frac{1 - C_u^2/C_I^2}{1 + C_u^2}\right), \tag{1}$$

with a 5×5 window, $C_u^2 = 1/\mathrm{ENL}$ ($\mathrm{ENL} = 4.4$ for IW GRD) and $C_I^2 = \mathrm{Var}(x)/\bar{x}^2$.

Monthly compositing. The per-pixel median of all filtered scenes in a month forms the composite, which also acts as temporal multi-looking. It is converted back to dB, and the VV−VH difference is added as a third channel.

C. Labels and their matching to the imagery

Labels come from Dynamic World V1 [1], which provides per-image class probabilities for nine classes from Sentinel-2. For each area and month the probabilities of all cloud-free observations were averaged and the most probable class was taken, on exactly the same 10 m grid as the radar composite, so image and label refer to the same month and footprint. Pixels whose mean top-class probability is below 0.3, or that had no cloud-free observation, are set to an ignore index. In a preliminary dataset of German study areas, a threshold of 0.5 had removed almost all grassland and bare-ground pixels (0.7 % and 0.05 % of the labels remained), so 0.3 was used. Grass and shrub/scrub were merged into *rangeland*, which gives eight classes identical to the Esri map without its cloud class (Table III); this choice allows a class-by-class comparison with that independent map in the case study.

TABLE II
Preprocessing chain

Step	Implementation
Calibration, terrain correction, dB	Earth Engine S1_GRD: orbit file, border and thermal noise removal, σ^0, SRTM/ASTER terrain correction
Scene selection	IW, VV+VH, 10 m, one orbit pass per area
Border noise	VV < -30 dB masked
Speckle filter	Lee, 5×5, ENL 4.4, linear power, per scene
Compositing	monthly per-pixel median; VV, VH, VV−VH in dB
Label matching	Dynamic World, same month and grid: mean class probability → argmax
Label quality	top-class probability < 0.3 or no observation → ignored
Patching	256×256, non-overlapping; drop patches with S1 gaps or < 50 % labels
Normalisation	clip VV $[-25, 0]$, VH $[-32, -5]$, VV−VH $[-5, 20]$ dB; z-score (training statistics)
Augmentation	flips, $90°$ rotations, ± 1 dB brightness (VV and VH jointly), speckle noise $\sigma = 0.3$ dB

TABLE III
Classes and pixel distribution (% of labelled pixels)

Class	Train	Val	Test	Test temp.	Loss w_c
Water	9.10	16.05	5.20	9.83	1.30
Trees	11.49	12.73	13.97	11.82	1.03
Flooded vegetation	2.85	2.54	0.08	1.55	4.14
Crops	20.26	23.86	12.88	20.05	0.58
Rangeland*	12.12	4.36	11.08	12.97	0.97
Built-up	23.65	24.70	24.20	23.37	0.50
Bare	17.97	15.74	19.06	17.51	0.66
Snow/ice	2.56	0.02	13.54	2.90	4.62

*Dynamic World grass and shrub/scrub merged.

D. Dataset

The export downloaded, for each area-month, four 1024×1024 tiles with six bands (VV, VH, VV−VH, Dynamic World class, confidence, number of observations) through Earth Engine's computePixels interface. In the final export run over all 633 area-months, 511 were complete, 40 were skipped because no Sentinel-1 scene or no cloud-free label existed (e.g., Amazon and Lagos in cloudy months), and 82 could not be

completed after the project had used up its monthly noncommercial quota. This removed all months of the Everest area, so snow/ice in training comes only from the Aletsch glacier, whereas the test set contains a large glacier (Vatnajökull). In total 2 265 tiles were downloaded and cut into the patches of Table I. The dataset is available on Kaggle (see Data Availability).

IV. Methods

A. Models

Three architectures of increasing capacity were compared first (Table IV): a *baseline CNN* of four 3×3 convolutions at full resolution (receptive field 90 m), which tests how much backscatter and local texture alone explain; a *U-Net* [10] trained from scratch with four downsampling stages (receptive field of about 2 km) and spatial dropout; and *DeepLabV3+* [13] with an ImageNet-pretrained ResNet50 cut after the 1/16-resolution stage, ASPP (rates 6, 12, 18 plus image pooling) and a decoder fusing 1/4-resolution features. The three SAR channels replace RGB; low-level pretrained filters transfer and are fine-tuned.

Six DeepLabV3+ variants then tested the encoder: ResNet101, ResNet152 and Xception, each in the standard cut-off form and as an *atrous encoder* (AE) that keeps the full backbone but replaces the stride of its last stage by dilation 2, so that features stay at output stride 16 as in the original DeepLab design [12]. Finally, a *multi-temporal* DeepLabV3+ with ResNet50 receives the target month and the two previous exported months (2–3 months apart) as nine channels; its first-layer filters are the pretrained filters copied to every month and divided by three, so that three identical months reproduce the single-month response.

B. Training

Normalisation and augmentation. Each channel is clipped to a fixed dB range and z-scored with the mean and standard deviation of the training split only (Table II), and the same statistics are used at inference. During training, patches are randomly flipped and rotated by multiples of 90°, because land cover has no preferred orientation. In addition, the same random offset of up to ± 1 dB is added to VV and VH, which leaves their ratio unchanged and mimics calibration and incidence-angle differences between scenes, orbits and satellites (Sentinel-1A in the training period, Sentinel-1A and 1C in the test period), and Gaussian noise with $\sigma = 0.3$ dB imitates residual speckle. Larger or per-channel intensity changes were avoided because they would alter the physical meaning of the backscatter, e.g., make water resemble vegetation.

Loss and optimisation. All models minimise class-weighted cross-entropy plus soft Dice over labelled pixels p,

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}, \qquad \mathcal{L}_{\text{CE}} = -\frac{\sum_p w_{y_p} \log s_{p,y_p}}{\sum_p w_{y_p}}, \tag{2}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{K} \sum_{c=1}^{K} \frac{2 \sum_p g_{pc} s_{pc} + 1}{\sum_p (g_{pc} + s_{pc}) + 1}, \tag{3}$$

where s are softmax scores, g one-hot labels and $w_c = \text{clip}_{[0.2,10]}\big(\text{median}(f)/f_c\big)$ the median-frequency weights [20] computed from the class frequencies f_c (Table III); Dice [21] measures overlap per class and is less dominated by frequent classes. Optimisation used AdamW [22] (weight decay 10^{-4}) with mixed precision on an NVIDIA RTX A6000 in TensorFlow 2.20. After every epoch, exact mIoU and overall accuracy were computed on the validation set and on 1 000 fixed training patches. The learning rate was halved after 3 epochs without validation-mIoU improvement, training stopped after 8 such epochs (at most 50) and the best epoch was kept. A model was labelled *underfitting* if its training mIoU at the best epoch was below 0.35 and *overfitting* if training exceeded validation mIoU by more than 0.15.

TABLE IV
Model overview

Model	Input	Params	Encoder	LR	Batch
Baseline CNN	1 month	0.08 M	– (4 conv layers)	10^{-3}	16
U-Net	1 month	7.9 M	from scratch	10^{-3}	16
DeepLabV3+	1 month	11.9 M	ResNet50 (ImageNet), cut at 1/16	$3 \cdot 10^{-4}$	16
DeepLabV3+ (E1–E3)	1 month	24.6–52.6 M	Xception, ResNet101, ResNet152	$3 \cdot 10^{-4}$	8
AE-DeepLabV3+ (E4–E6)	1 month	37.7–75.2 M	same, full backbone, atrous last stage	$3 \cdot 10^{-4}$	8
DeepLabV3+ (multi-temporal)	3 months	11.9 M	ResNet50, first layer for 9 channels	$3 \cdot 10^{-4}$	16

C. Evaluation

With confusion matrix n_{ij} (reference i, prediction j) over N labelled test pixels,

$$\mathrm{OA} = \frac{1}{N}\sum_{c} n_{cc}, \qquad \mathrm{IoU}_c = \frac{n_{cc}}{n_{c\cdot} + n_{\cdot c} - n_{cc}}, \tag{4}$$

$$F1_c = \frac{2n_{cc}}{n_{c\cdot} + n_{\cdot c}}, \tag{5}$$

and mIoU and mean F1 average over classes present in the reference. Model selection used the validation mIoU only; differences smaller than its epoch-to-epoch variation were treated as ties in favour of the simpler model.

V. Results

A. Architecture comparison

Table V and Fig. 2 summarise the first comparison. Context and pretraining help: test mIoU rises from 0.203 (baseline) to 0.275 (U-Net) and 0.304 (DeepLabV3+), which was selected on validation mIoU. The baseline is weak even on training data (underfitting), whereas U-Net and DeepLabV3+ reach a training mIoU near 0.6 but only about 0.4 on the validation regions (overfitting). Early stopping ended the runs after 18–25 epochs.

B. Generalisation to new regions and dates

On the training regions in 2025–26, U-Net and DeepLabV3+ reach OA 0.76–0.78 and mIoU 0.56–0.58 (Table V), against 0.50–0.51 and 0.27–0.30 on the unseen test regions. The models therefore cope well with a later period but much less with new landscapes: the overfitting is geographic. Per region (Table VI), Gabon, Mexico City, Pantanal and Dhaka are classified with OA 0.74–0.97, while Atacama (salt flats, OA 0.03) and Vatnajökull (glacier, OA 0.00) fail almost completely. The confusion matrix (Fig. 3) explains the failure: 66 % of the bare pixels are predicted as snow/ice and 40 % of the snow/ice pixels as water. Crops and rangeland are confused in both directions (22 % and 27 %), and flooded vegetation is mostly predicted as trees or water. Built-up (IoU 0.78), trees (0.54) and water (0.36) are the most reliable classes (Table VII).

C. Encoder experiments

Because both deep models overfit, the encoder was varied in a controlled experiment with otherwise identical settings (Table VIII, Fig. 4). Deeper encoders fit the training regions better without generalising better: ResNet152 has the best validation mIoU (0.410) but a test mIoU of only 0.257. Xception overfits least (gap 0.14) but also learns least. The atrous encoder lowers the validation mIoU but raises the test mIoU for all three backbones (+0.008, +0.042, +0.025) and is the only change that lifts snow/ice above zero (IoU 0.10 for AE-ResNet152). None of the six models with 25–75 M parameters exceeds the 11.9 M-parameter ResNet50 (0.304), and validation and test rankings disagree. With four validation regions, a single region can decide the ranking, so the validation lead of ResNet152 (+0.011) was treated as a tie and the simpler ResNet50 was kept.

D. Multi-temporal input

The remaining confusions concern classes that differ mainly in their seasonal behaviour, so

TABLE V
Architecture comparison. Test: nine unseen regions, 2025–26. Test temporal: training regions, 2025–26

			mIoU at best epoch			Test			Test temporal	
Model	Epochs	Best	Train	Val	Gap	OA	mIoU	mF1	OA	mIoU
Baseline CNN	18	10	0.337	0.306	0.031	0.420	0.203	0.303	0.557	0.332
U-Net	25	17	0.593	0.381	0.212	0.496	0.275	0.365	**0.782**	**0.576**
DeepLabV3+ (ResNet50)	24	16	0.619	**0.399**	0.220	**0.510**	**0.304**	**0.402**	0.755	0.565

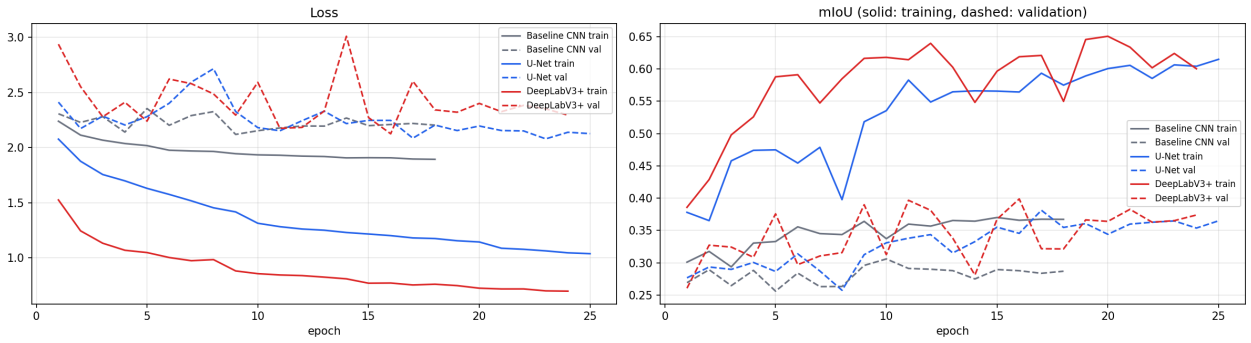


Fig. 2. Training curves of the three architectures (solid: training, dashed: validation). Validation mIoU fluctuates by up to ± 0.08 between epochs because the four validation regions differ strongly.

TABLE VI
DeepLabV3+ (ResNet50) per test region

Region (landscape)	OA	mIoU	Patches
Gabon (forest)	0.973	0.139	84
Mexico City (urban)	0.868	0.199	593
Pantanal (wetland)	0.770	0.160	642
Dhaka (urban)	0.742	0.262	359
Dubai (urban, desert)	0.650	0.198	574
Kansas (cropland)	0.605	0.283	615
Canterbury NZ (cropland)	0.452	0.144	509
Atacama (salt flats)	0.027	0.004	641
Vatnajökull (glacier)	0.004	0.004	519

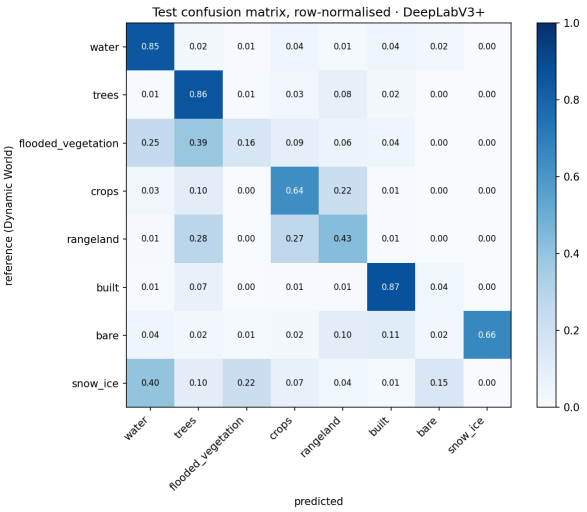


Fig. 3. Row-normalised test confusion matrix of DeepLabV3+ (ResNet50).

the third step changed only the input to three months. Both models were evaluated on the same patches that have two earlier months (Table IX); this reduces the data to 11 283 training, 810 validation and 2 814 test patches. Temporal information helps exactly where expected: cropland IoU rises from 0.552 to 0.646 and the two cropland test regions improve (Kansas $+7.9$ pp OA, Canterbury $+4.5$ pp). Overall, however, the model overfits more (gap 0.36 against 0.22), trees (-0.195 IoU), rangeland (-0.115) and the Pantanal wetland (-8.6 pp OA) get worse, and the test mIoU falls from 0.313 to 0.276. With three times more information per pixel, the network learns the seasonal rhythm of the training regions, which does not transfer to other climates and hemispheres; only the crop

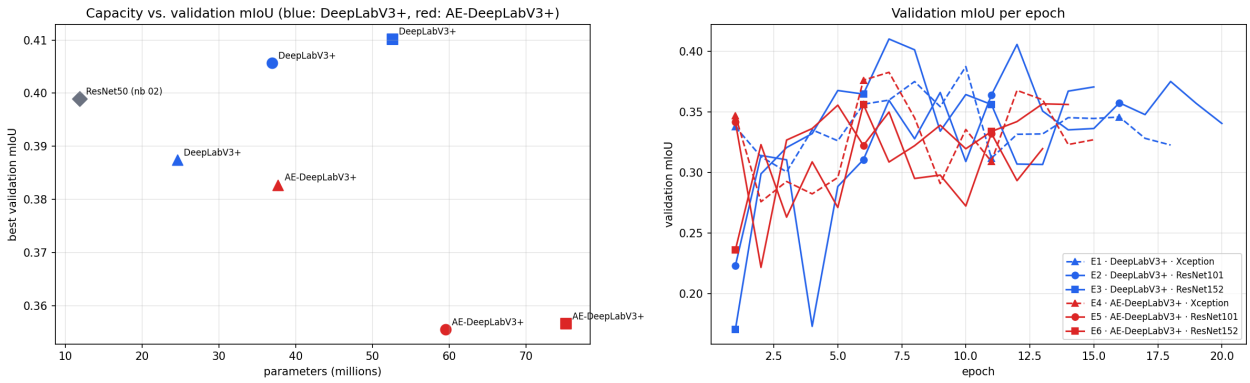


Fig. 4. Encoder experiments: capacity versus best validation mIoU (left) and validation mIoU per epoch (right). Blue: DeepLabV3+, red: AE-DeepLabV3+.

TABLE VII
Test IoU / F1 per class

Class	Baseline	U-Net	DeepLabV3+
Water	0.28 / 0.44	0.33 / 0.50	**0.36** / **0.53**
Trees	0.34 / 0.51	0.33 / 0.50	**0.54** / **0.70**
Flooded veg.	0.01 / 0.01	0.00 / 0.00	0.00 / 0.01
Crops	0.21 / 0.34	**0.51** / **0.67**	0.46 / 0.63
Rangeland	**0.27** / **0.42**	0.18 / 0.30	**0.27** / **0.42**
Built-up	0.52 / 0.68	**0.82** / **0.90**	0.78 / 0.88
Bare	0.01 / 0.02	0.01 / 0.02	0.02 / 0.04
Snow/ice	0.00 / 0.00	0.01 / 0.03	0.00 / 0.00

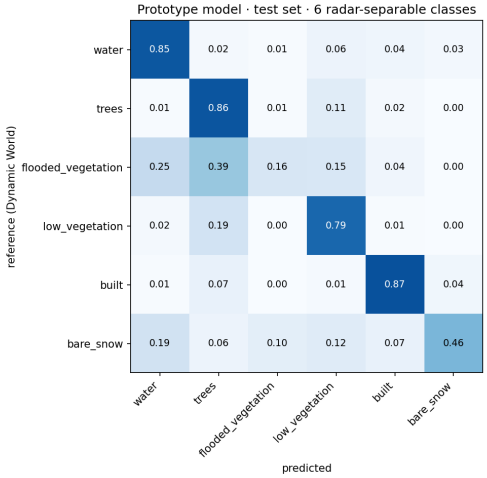


Fig. 5. Test confusion matrix of the prototype model in the six-class scheme.

cycle is similar enough. The validation difference of $+0.005$ is again within the noise, so the single-month model remains the prototype model.

E. Radar-separable class scheme

All ten models end near the same test mIoU, and their errors concentrate in the two class pairs identified above. These pairs are easy to separate in optical imagery, from which the eight classes originate, but not in C-band backscatter: dry snow is nearly transparent [5], [9] and resembles the ground beneath, and fields, grassland and shrubland have similar roughness and moisture in a single month. The predictions of the prototype model were therefore re-evaluated, without retraining, after merging crops and rangeland into *low vegetation* and bare and snow/ice into *bare/snow* (Table X, Fig. 5). The overall accuracy rises from 0.510 to 0.715 and the mIoU from 0.304 to 0.461; bare/snow reaches IoU 0.45 (previously 0.02 and 0.00) and low vegetation 0.63. Without the two extreme test regions, the six-class OA is 0.772, close to the 0.76–0.78 reached on the training regions. Flooded vegetation, with only 0.08 % of the test pixels, stays near zero but counts as a full class in the mean; the other five classes average an IoU of about 0.55.

VI. Prototype

A. System architecture

The prototype (Fig. 6) is a Streamlit web application on top of a Python package that is shared with the training code, so that inference uses exactly the training preprocessing. The user

TABLE VIII
Encoder experiments (batch 8). Gap = train − val mIoU

DeepLabV3+ encoder	Parameters	Train mIoU	Val mIoU	Gap	Test OA	Test mIoU
E1 Xception	24.6 M	0.528	0.387	0.140	0.431	0.245
E2 ResNet101	36.9 M	0.626	0.406	0.221	0.496	0.289
E3 ResNet152	52.6 M	0.616	**0.410**	0.206	0.468	0.257
E4 AE Xception	37.7 M	0.593	0.383	0.211	0.487	0.270
E5 AE ResNet101	59.5 M	0.579	0.356	0.223	0.511	0.297
E6 AE ResNet152	75.2 M	0.614	0.357	0.257	**0.518**	**0.299**
Reference: ResNet50 (batch 16)	11.9 M	0.619	0.399	0.220	0.510	0.304

TABLE IX
Single-month vs. three-month input on the same patches

	Validation		Test		Test IoU	
Input	OA	mIoU	OA	mIoU	crops	trees
1 month	0.615	0.344	**0.487**	**0.313**	0.552	**0.517**
3 months	**0.637**	**0.348**	0.480	0.276	**0.646**	0.322

TABLE X
Same predictions (prototype model) under two class schemes

Test regions	Classes	OA	mIoU	mF1
All nine	8 (original)	0.510	0.304	0.402
	6 (merged)	**0.715**	**0.461**	**0.585**
Without the two extremes*	8 (original)	0.692	0.355	0.454
	6 (merged)	**0.772**	0.459	0.556

*Atacama (salt flats) and Vatnajökull (glacier).

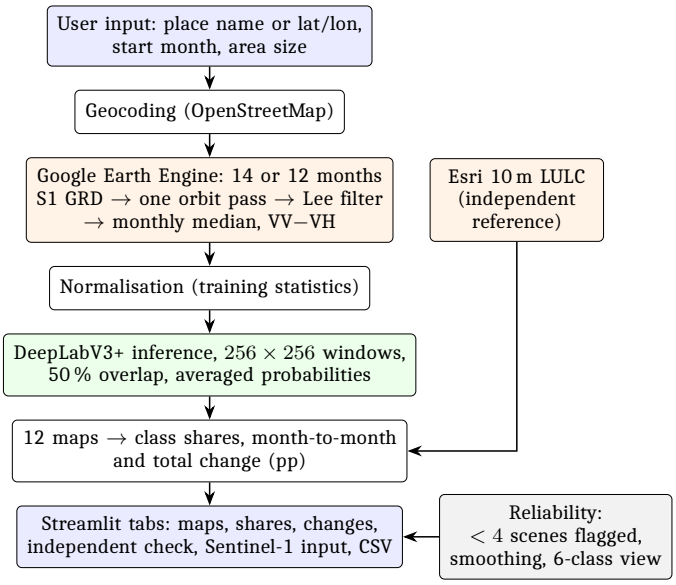


Fig. 6. Prototype architecture.

enters a place name (resolved through OpenStreetMap Nominatim, with a choice among up to five matches) or coordinates, a start month and an area of 2.56, 5.12 or 10.24 km. A run takes about 30–90 s on a laptop CPU.

B. Data retrieval and monthly compositing

For the selected square, Earth Engine selects the orbit pass with more scenes over the whole twelve months and uses it for every month, then builds each monthly composite with the training chain (Table II); a model with temporal input additionally receives the earlier months. Composites are requested on the exact 10 m UTM grid with `computePixels`, so no export or storage is needed. Months without any scene are shown as empty and excluded from the statistics, and months composited from fewer than four scenes are flagged because they retain more speckle and are dominated by single acquisitions.

C. Class percentages and changes

For month m with valid (radar-covered) pixels $V(m)$ and predicted class $\hat{y}_p(m)$, the class share and the changes in percentage points are

$$s_c(m) = 100 \cdot \frac{|\{p \in V(m) : \hat{y}_p(m) = c\}|}{|V(m)|}, \quad (6)$$

$$\Delta_c(m) = s_c(m) - s_c(m-1), \quad (7)$$

$$\Delta_c^{\text{tot}} = s_c(m_{12}) - s_c(m_1). \quad (8)$$

Two optional views help to judge the result: a three-month majority filter per pixel that

suppresses one-month flicker, and the six-class scheme of Table X. For the independent check, the most frequent class per pixel over the twelve months is compared with the Esri 10 m Annual LULC map [2], which is produced from Sentinel-2 by a different model and was not used in training.

D. Case study: Munich, 2025

Munich (10.24 km around the city centre, January–December 2025, 7–14 descending scenes per month) lies 260 km from the nearest study area. The twelve maps (Fig. 7) reproduce the Isar, the Englischer Garten and the riverside forests. The pixel agreement with Esri 2025 is 87.9 %; built-up, which covers 92 % of the area, reaches F1 0.94 against Esri (Fig. 8). The largest difference is trees (model 14.9 %, Esri 4.6 %), largely a difference of definitions: Esri maps trees inside the city as built area, whereas Dynamic World, the training reference, maps them as trees.

The monthly shares (Fig. 9) show a seasonal cycle without real land-use change: trees rise from 10.6 % in January to 17.1 % in July and built-up falls from 86.7 % to 81.1 %, because leafless trees produce little volume scattering. The total change January→December is $+5.3$ pp trees and -4.5 pp built-up, which would correspond to 4.7 km^2 of lost built-up area. One genuine temporary change is captured: a large smooth square, most likely the Theresienwiese, appears as water from January to June (specular reflection) and as built-up from July to November, when the Oktoberfest structures are put up, used and removed.

VII. Discussion

A. Regions unlike the training data

The gap between familiar regions (mIoU 0.56–0.58) and unseen regions (about 0.30) is the dominant error source, and neither larger encoders nor temporal input reduced it. Landscapes absent from training fail: the hyper-arid Atacama salt flats and the Vatnajökull ice cap. The quota-related loss of the Everest area made this worse by leaving a single glacier in training. The prototype shows the same effect for Kampala, where no East African city exists in training. Training on many more climates, and validation with more regions or region-wise cross-validation, would address both generalisation and the unstable model selection observed here.

B. SAR-specific confusion

Bare ground and snow/ice, cropland and rangeland, and smooth paved surfaces and water are confused because their C-band backscatter is similar; flooded vegetation is labelled as trees or water depending on whether canopy or open water dominates the signal. These are physical limits of a single-frequency, dual-polarisation sensor rather than deficits of a particular network, which is why the radar-separable scheme improves the accuracy so strongly. A Sentinel-1-only product should report such merged classes, or at least communicate the expected confusions.

C. Seasonal effects that resemble land-use change

Leaf-on/leaf-off, crop growth and harvest, soil moisture, flooding and snow change the backscatter within a year and move pixels across class boundaries. In Munich, the scene-average backscatter varied by less than 1 dB, yet the tree share changed by 6.5 pp. A further non-physical change is the number of acquisitions: in Munich the scenes per month rose from 8 to 14 during 2025, largely because Sentinel-1C (launched in December 2024) joined the constellation, and composites from two or three scenes (Kampala in early 2025, Delhi throughout 2025) produced swings of ± 10–25 pp. Changes should therefore be compared between the same months of different years, and low-scene months interpreted with care.

D. How classification errors inflate change

If a pixel were classified correctly with probability p in every month and errors were independent, an unchanged pixel would appear to change between two months with probability of about $1 - p^2$, i.e. 74 % for the test OA ($p = 0.51$) and 23 % for the Munich agreement ($p = 0.879$). In Munich only 4.5 % of the pixels change class

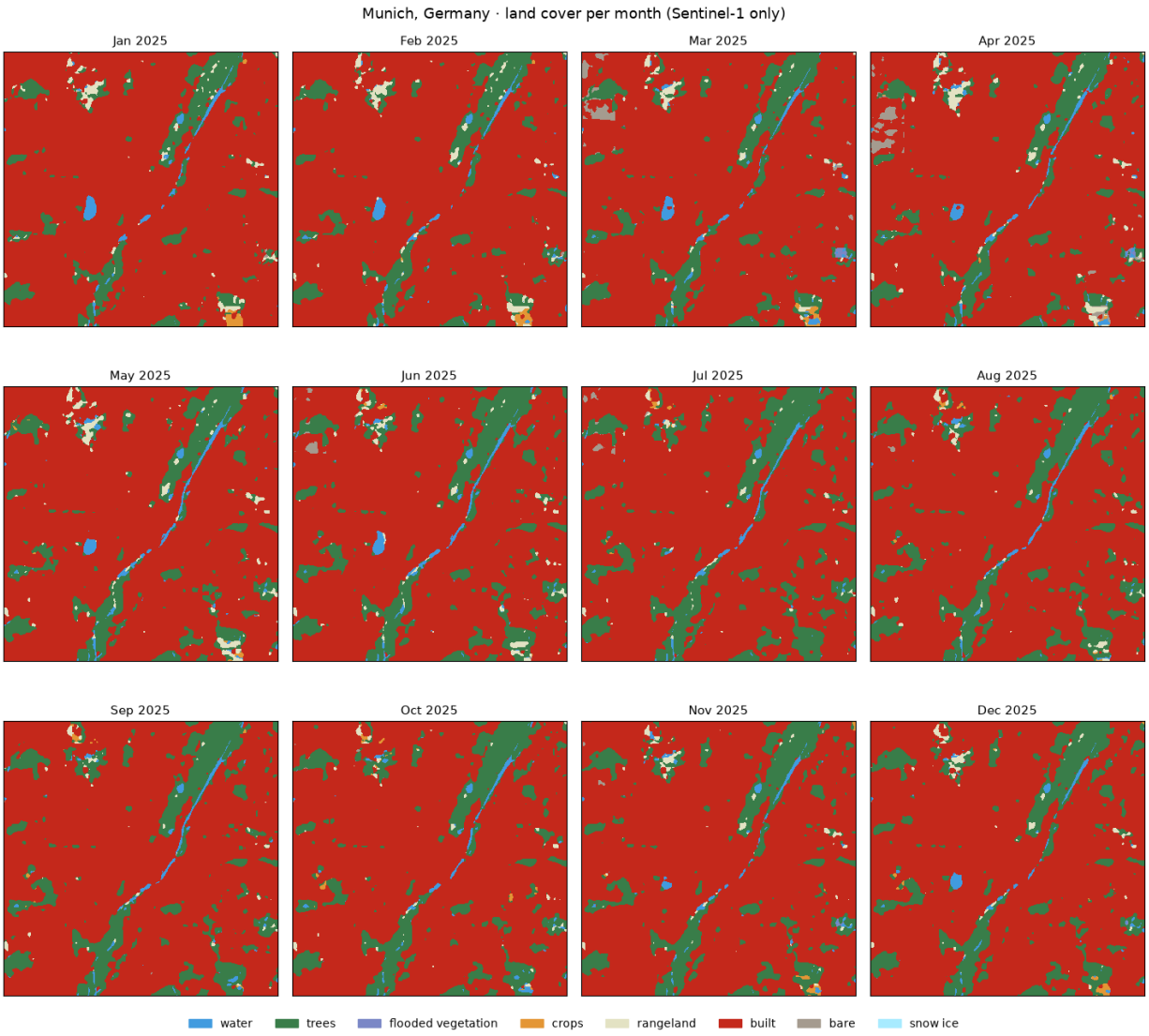


Fig. 7. Case study Munich: twelve monthly land-cover maps (2025) from Sentinel-1 only.

from one month to the next: most errors are systematic, repeat every month, and bias the class shares (trees $+10$ pp against Esri) more than the changes. The part that does vary, mostly seasonal, inflates the net change: the three-month majority filter reduces the share of changing pixels from 4.5 % to 1.9 %, but the total change only from $+5.3$ to $+4.1$ pp (trees) and -4.5 to -3.1 pp (built-up), because it removes random flicker but not systematic seasonal shifts. Net changes also hide gross changes, since pixels moving into and out of a class cancel.

E. Limitations of the prototype

Labels are an automatic product with its own errors and an optical class concept, which caps the attainable accuracy. The prototype processes 256×256 windows; because the ASPP image-pooling branch summarises each window, an uncertain window can switch class as a block, producing visible rectangles in low-scene months. Increasing the window overlap from 50 % to 75 % raised the agreement with Esri slightly (Delhi $+1.6$ pp, Kampala $+4.8$ pp) but did not reduce the monthly swings (17.9 vs. 17.6 pp for built-up in Delhi) and made runs almost four times slower, so the noise originates in the input

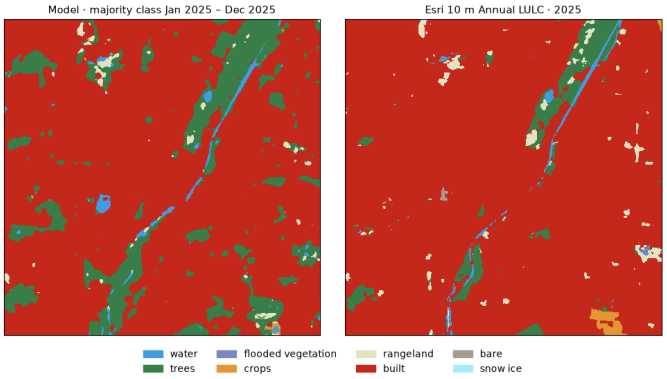


Fig. 8. Munich: majority class over 2025 (left) and the independent Esri 10 m Annual LULC map 2025 (right).

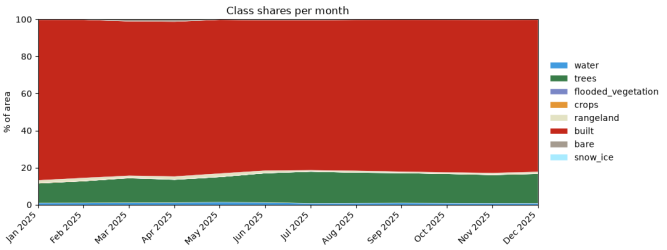


Fig. 9. Munich: class shares per month. The tree share follows the leaf-on/leaf-off cycle.

composites rather than at window seams. The Esri reference is annual and currently ends in 2025, and Earth Engine quotas limit the number of runs.

VIII. Conclusion

Sentinel-1 alone supports LULC segmentation that is reliable for built-up areas, forests and water and that works independently of clouds, but its accuracy on unseen landscapes is limited by two factors that a larger network does not remove: the geographic diversity of the training data and an optical class scheme that demands distinctions C-band radar cannot make. Ten models trained under one protocol reached the same ceiling of about 0.30 test mIoU, while the same predictions reach an overall accuracy of 0.72 in a radar-separable six-class scheme and 0.77 without the two landscapes absent from training. For monitoring, systematic errors largely cancel in monthly changes, whereas seasonal backscatter changes and acquisition density do not; the prototype therefore reports reliability alongside the statistics. Future work should add training regions from missing climates and landscapes, evaluate with more validation regions, and combine temporal input with that broader data.

Data and Code Availability

Dataset (Earth Engine tiles and training patches, about 25 GB): https://www.kaggle.com/datasets/namratabhoyar/sentinel1-lulc-global. Code, notebooks with all outputs, trained weights and prototype: https://github.com/namrata1104/lulc-s1.

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